

Week 1 - Builder Journal

Submitted By: Muhammad Hassan

Why did you choose Computer Vision:

I chose Computer Vision because it is one of the most impactful fields within Artificial Intelligence. Unlike traditional software, computer vision enables machines to interpret and understand images and videos in a way that supports intelligent decision-making. Since visual information is generated everywhere from mobile phones and surveillance cameras to medical scanners and autonomous vehicles I believe this field has enormous potential to solve practical problems.

As a Artificial Intelligence student, I want to build systems that combine programming, machine learning, and image processing to create meaningful solutions. Computer Vision offers the opportunity to work on challenging real-world problems while continuously learning modern AI technologies. I also believe that expertise in this field will be valuable for future research, internships, and professional opportunities.

What applications interest you most:

The areas that interest me the most include medical image analysis, autonomous driving, industrial quality inspection, intelligent surveillance, agriculture, and robotics. In healthcare, Computer Vision can assist doctors by identifying diseases from medical images with greater speed and consistency. In manufacturing, it can automatically inspect products and detect defects that may be difficult to identify manually.

I am also interested in object detection and tracking because these technologies are used in smart traffic systems, security monitoring, retail analytics, and warehouse automation. My long-term goal is to develop AI applications that improve efficiency, safety, and decision-making while addressing practical challenges faced by industries and society.

What did you learn this week:

During this week, I learned the fundamentals of digital image processing using OpenCV. I explored how images can be represented in different color spaces such as RGB, Grayscale, and HSV, and why different representations are useful for different computer vision tasks. I also studied image filtering techniques, including Average Blur, Gaussian Blur, and Median Blur, and understood how they reduce noise before further processing.

In addition, I learned edge detection using the Canny algorithm, contour detection for identifying object boundaries, and shape detection

through contour approximation. These practical experiments helped me understand that preprocessing plays a vital role in improving the quality of images before applying advanced AI models. Overall, the week's activities strengthened my understanding of the basic building blocks of Computer Vision.

Biggest challenge:

The biggest challenge I faced was understanding how different parameter values affect the output of image processing algorithms. For example, selecting suitable kernel sizes for image filtering and choosing appropriate thresholds for edge detection required experimentation because small adjustments produced noticeably different results.

Another challenge was understanding the purpose of each preprocessing technique and determining when one method should be preferred over another.

How did you solve it:

I addressed these challenges by experimenting with different parameter values and carefully comparing the resulting outputs. Instead of relying on a single example, I tested the algorithms on multiple images to observe how the results changed under different conditions. I also referred to the OpenCV documentation and reviewed the theory behind each algorithm to understand the reasoning behind the parameters.

This combination of practical experimentation and theoretical study improved my understanding significantly. By repeating the experiments and analyzing the results, I became more confident in selecting suitable techniques and parameters for different image processing tasks.

Goals for Week 2:

- During Week 2, my goal is to transition from traditional image processing to modern AI-powered Computer Vision by understanding the complete lifecycle of an object detection project. I plan to learn how to collect a high-quality dataset, perform accurate image annotation, and prepare the dataset for training in a format suitable for deep learning models.

Another important objective is to successfully train a custom YOLO object detection model and evaluate its performance using appropriate metrics. I also want to perform inference on both images and videos to verify that the trained model can detect objects reliably in different scenarios.

Finally, I aim to deploy a simple application capable of running the trained model for real-time object detection. By the end of Week 2, I expect to have a clear understanding of the end-to-end workflow involved in developing modern Computer Vision applications, from data collection to deployment, while improving my practical implementation skills and confidence in building AI-based solutions independently.